# ☕Cafe Order & Billing System

Python Console Application — Project Documentation

Language: Python 3

## 1. Project Description

The Cafe Order & Billing System is a Python console application that lets a customer register their details, browse a menu, order one or more items, and instantly receive a complete, itemized receipt. It validates all user input, calculates the order subtotal, applies tax automatically based on the chosen payment method, and displays the final payable amount along with the current date and time.

## 2. Features

- Customer name validation — alphabets only, spaces auto-removed, auto-capitalized
- Phone number validation — must be exactly 11 digits
- Dynamic menu display with live pricing (7 items)
- Multi-item ordering loop — customer can order as many items as they like
- Quantity validation — only positive whole numbers accepted
- Payment method selection — Cash or Card
- Automatic tax calculation — 5% for card payments, 16% for cash payments
- Auto-generated itemized receipt with date, time, subtotal, tax, and grand total

## 3. Menu & Pricing

| Item | Price |
|---|---|
| Cookie | Rs 200 |
| Chocolate Latte | Rs 300 |
| Sandwich | Rs 100 |
| Coffee | Rs 90 |
| Cake Slice | Rs 300 |
| Brownie | Rs 100 |
| Hot Chocolate | Rs 400 |

## 5. Code Execution

The program was executed cell-by-cell in Google Colab using a sample customer order. Each screenshot below shows the executed code cell together with its live output, confirming the program runs correctly end-to-end.

### Cell 1 — Imports, menu dictionary, and welcome message

C cafe_order_system.ipynb â€” Google Colab

[1]

```
from datetime import datetime

menu = {
    &#x27;cookie': 200,
    &#x27;chocolate_latte': 300,
    &#x27;sandwich': 100,
    &#x27;coffee': 90,
    &#x27;cake slice': 300,
    &#x27;brownie': 100,
    &#x27;hot chocolate':400,
}

print("-------- Welcome to Our Cafe --------")
```

```
-------- Welcome to Our Cafe --------
```

### Cell 2 — Customer name and phone number validation

cafe_order_system.ipynb â€” Google Colab

```
# Customer Name Validation
while True:
    name = input("Enter customer name: ").replace(" ", "")
    name = name.capitalize()
    if name.isalpha():
        break
    else:
        print("Name should contain only alphabets.")

# Phone Number Validation
while True:
    phone = input("Enter phone number (11 digits): ")
    if phone.isdigit() and len(phone) == 11:
        break
    else:
        print("Phone number must be exactly 11 digits.")
```

```
Enter customer name: Ali
Enter phone number (11 digits): 03001234567
```

## Cell 3 — Menu display

cafe_order_system.ipynb â€” Google Colab

```
# Display Menu
print("\n--- Menu ---")
for item, price in menu.items():
    print(f"{item} : Rs{price}")
```

```
--- Menu ---
cookie : Rs200
chocolate_latte : Rs300
sandwich : Rs100
coffee : Rs90
cake slice : Rs300
brownie : Rs100
hot chocolate : Rs400
```

## Cell 4 — Item ordering loop (sample order: 2x Coffee, 1x Cake Slice)

C cafe_order_system.ipynb â€” Google Colab

[4]

```
order_total = 0
ordered_items = []

# Ordering Items
while True:
    item = input("\nWhat item do you want to order? = ").lower()
    if item in menu:
        while True:
            qty = input("Enter quantity: ")
            if qty.isdigit() and int(qty) > 0:
                qty = int(qty)
                break
            else:
                print("Enter valid quantity.")
        cost = menu[item] * qty
        order_total += cost
        ordered_items.append((item, qty, cost))
        print("Item added successfully.")
    else:
        print("Item not available.")

    another = input("Do you want to order another item? (yes/no): ").lower()
    if another != "yes":
        break
```

```
What item do you want to order? = coffee
Enter quantity: 2
Item added successfully.
Do you want to order another item? (yes/no): yes

What item do you want to order? = cake slice
Enter quantity: 1
Item added successfully.
Do you want to order another item? (yes/no): no
```

## Cell 5 — Payment method selection and tax calculation

C cafe_order_system.ipynb â€” Google Colab

[5]

```
# Payment Method
while True:
    payment_method = input("\nPayment method (cash/card): ").lower()
    if payment_method in ["cash", "card"]:
        break
    else:
        print("Invalid payment method.")

# Tax Calculation
if payment_method == "card":
    tax_rate = 0.05   # 5%
else:
    tax_rate = 0.16   # 16%

tax_amount = order_total * tax_rate
grand_total = order_total + tax_amount

# Date & Time
now = datetime.now()
date = now.strftime("%d-%m-%Y")
time = now.strftime("%I:%M %p")
```

```
Payment method (cash/card): card
```

## Cell 6 — Final receipt generation

[6]

```
# Receipt
print("\n========== RECEIPT ==========")
print("Date :", date)
print("Time :", time)
print("Customer Name :", name)
print("Phone Number  :", phone)
print("Payment Method:", payment_method.upper())
print("\nItems Ordered:")
print("-----------------------------")
for item, qty, cost in ordered_items:
    print(f"{item} x{qty} = Rs{cost}")
print("-----------------------------")
print("Subtotal        : Rs", order_total)
print(f"Tax ({int(tax_rate*100)}%)       : Rs", tax_amount)
print("Total Payable   : Rs", grand_total)
print("=============================")
print("Thank you for visiting ")
```

```
========== RECEIPT ==========
Date : 06-07-2026
Time : 09:20 AM
Customer Name : Ali
Phone Number  : 03001234567
Payment Method: CARD

Items Ordered:
-----------------------------
coffee x2 = Rs180
cake slice x1 = Rs300
-----------------------------
Subtotal        : Rs 480
Tax (5%)       : Rs 24.0
Total Payable   : Rs 504.0
=============================
Thank you for visiting
```

## 6. Sample Output — Final Receipt

Sample run — Customer: Ali | Phone: 03001234567 | Order: 2x Coffee, 1x Cake Slice | Payment: Card

```
========== RECEIPT ==========
Date : 06-07-2026
Time : 09:20 AM
Customer Name : Ali
Phone Number  : 03001234567
Payment Method: CARD

Items Ordered:
-----------------------------
coffee x2 = Rs180
cake slice x1 = Rs300
-----------------------------
Subtotal        : Rs 480
Tax (5%)       : Rs 24.0
Total Payable   : Rs 504.0
=============================
Thank you for visiting
```

## 7. Conclusion

The Cafe Order & Billing System was implemented, executed, and verified successfully in Google Colab. All input validations, the ordering loop, tax calculation logic, and the final receipt generation worked as intended, producing accurate and correctly formatted output for the sample order tested.